

Shiny and renv

How to develop Shiny applications

Milan, 9th December 2020



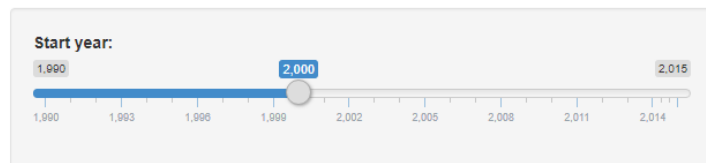


A Shiny app and its issues

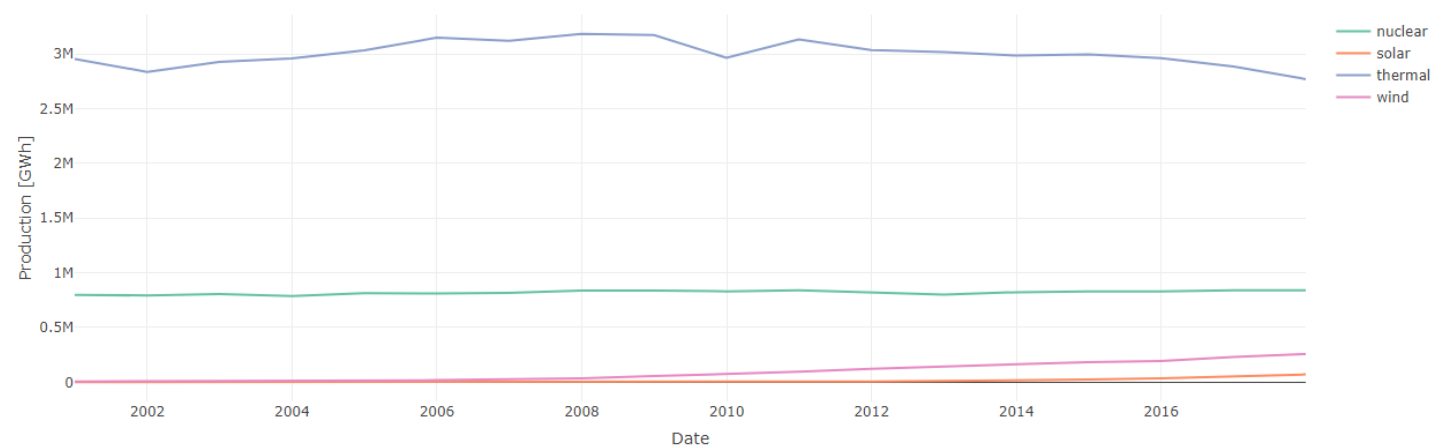


The app

Production of electric energy in the USA



Energy production by year



Summary statistics

date	nuclear	solar	thermal	wind
Min. :2000-12-31	Min. :787818	Min. : 709	Min. :2770050	Min. : 5650
1st Qu.:2005-04-01	1st Qu.:806071	1st Qu.: 1004	1st Qu.:2955504	1st Qu.: 15188
Median :2009-07-01	Median :821705	Median : 2304	Median :2990724	Median : 64961
Mean :2009-07-01	Mean :819483	Mean :12873	Mean :3004850	Mean : 91602
3rd Qu.:2013-09-30	3rd Qu.:835122	3rd Qu.:14440	3rd Qu.:3099529	3rd Qu.:162765
Max. :2017-12-31	Max. :839918	Max. :70980	Max. :3183440	Max. :257249



... and its problems

The screenshot displays the RStudio IDE interface. The main editor window shows the R script for a Shiny application named 'app.R'. The code defines a fluid page layout with a title panel, a sidebar panel containing a slider input for 'start_year', and a main panel with a plot and summary statistics. The file explorer on the right shows the project structure, including files like .gitignore, .Rhistory, app.R, electricity_production.csv, README.md, script.R, and shiny-bad-app.Rproj.

```
1 library(shiny)
2 library(Quandl)
3 library(tidyverse)
4 library(plotly)
5
6 ui <- fluidPage(
7   # Application title
8   titlePanel("Production of electric energy in the USA"),
9   # Sidebar with a slider input for the start year
10  sidebarLayout(
11    sidebarPanel(
12      sliderInput("start_year", "Start year:", min = 1990, max = 2015, value = 2000)
13    ),
14    # Show a plot of the evolution of energy production
15    mainPanel(
16      h2('Energy production by year'),
17      plotlyOutput("energy_plot"),
18      h2('Summary statistics'),
19      verbatimTextOutput('energy_summary')
20    )
21  )
22
23 # Define server logic required to draw the chart
24 server <- function(input, output, session) {
25   # ...
26 }
27
28 # Run the application
```

Environment History Connections Git Tutorial
Diff Commit Pull Push History More
Staged Status Path
New Branch master
Files Plots Packages Help Viewer
New Folder Delete Rename More
C: > Git > shiny-bad-app
Name Size Modified
.. 44 B Dec 7, 2020, 12:37 PM
.gitignore 16 B Dec 7, 2020, 4:16 PM
.Rhistory 1.7 KB Dec 7, 2020, 3:58 PM
app.R 1.1 KB Dec 7, 2020, 3:14 PM
electricity_production.csv 75 B Dec 7, 2020, 4:11 PM
README.md 718 B Dec 7, 2020, 3:43 PM
script.R 274 B Dec 8, 2020, 1:13 PM
shiny-bad-app.Rproj



Poor project structure

```
1 > fs::dir_tree()  
2 .  
3 +-- app.R  
4 +-- electricity_production.csv  
5 +-- README.md  
6 +-- script.R  
7 \-- shiny-bad-app.Rproj
```



Poor code design

```
1 server <- function(input, output, session) {
2
3   data_set <- reactive({
4     Quandl.api_key('9zz78YR1Y-iLD9Mh4CHq')
5     data_set <- Quandl::Quandl(c('UENG/EN_USA', 'UENG/ES_USA', 'UENG/ET_PROD_USA', 'UENG/EW_PROD_USA'))
6     #data_set <- readr::read_csv2('electricity_production.csv')
7     colnames(data_set) <- c('date', 'nuclear', 'solar', 'thermal', 'wind')
8     data_set_filtered <- data_set %>% filter(date > lubridate::ymd(paste0(input$start_year, '-01-01')))
9   })
10
11   long_data_set <- reactive(
12     pivot_longer(data_set(), !date, names_to = 'source', values_to = 'production')
13   )
14
15   output$energy_plot <- renderPlotly({
16     plot_ly(long_data_set(), x=~date, y=~production, color=~source) %>%
17       add_lines() %>%
18       layout(xaxis = list(title='Date'), yaxis = list(title='Production [GWh]'))
19   })
20
21   output$energy_summary <- renderPrint(
22     summary(data_set())
23   )
24 }
```

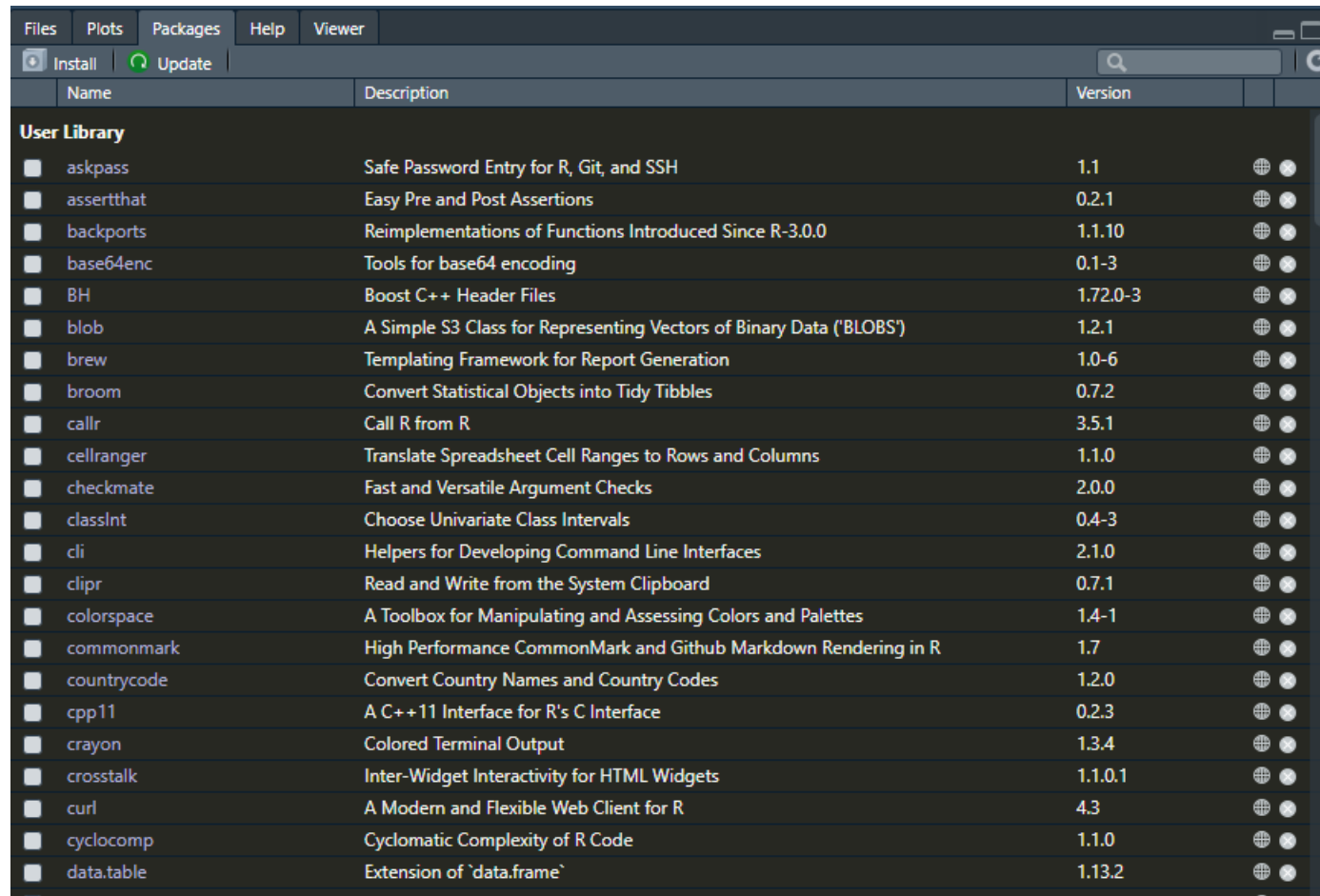


Poor code design

- No configuration
- No documentation
- No separation between application and business logic
- Too much reactivity
- Inconsistent styling
- No segregation of responsibilities



No dependency management



The screenshot shows the RStudio interface with the 'Packages' tab selected. The 'User Library' section displays a list of installed packages. Each row includes a checkbox, the package name, a description, the version number, and icons for installation, update, and removal.

	Name	Description	Version			
☐	askpass	Safe Password Entry for R, Git, and SSH	1.1	⊕	↻	⊗
☐	assertthat	Easy Pre and Post Assertions	0.2.1	⊕	↻	⊗
☐	backports	Reimplementations of Functions Introduced Since R-3.0.0	1.1.10	⊕	↻	⊗
☐	base64enc	Tools for base64 encoding	0.1-3	⊕	↻	⊗
☐	BH	Boost C++ Header Files	1.72.0-3	⊕	↻	⊗
☐	blob	A Simple S3 Class for Representing Vectors of Binary Data ('BLOBS')	1.2.1	⊕	↻	⊗
☐	brew	Templating Framework for Report Generation	1.0-6	⊕	↻	⊗
☐	broom	Convert Statistical Objects into Tidy Tibbles	0.7.2	⊕	↻	⊗
☐	callr	Call R from R	3.5.1	⊕	↻	⊗
☐	cellranger	Translate Spreadsheet Cell Ranges to Rows and Columns	1.1.0	⊕	↻	⊗
☐	checkmate	Fast and Versatile Argument Checks	2.0.0	⊕	↻	⊗
☐	classInt	Choose Univariate Class Intervals	0.4-3	⊕	↻	⊗
☐	cli	Helpers for Developing Command Line Interfaces	2.1.0	⊕	↻	⊗
☐	clipr	Read and Write from the System Clipboard	0.7.1	⊕	↻	⊗
☐	colorspace	A Toolbox for Manipulating and Assessing Colors and Palettes	1.4-1	⊕	↻	⊗
☐	commonmark	High Performance CommonMark and Github Markdown Rendering in R	1.7	⊕	↻	⊗
☐	countrycode	Convert Country Names and Country Codes	1.2.0	⊕	↻	⊗
☐	cxx11	A C++11 Interface for R's C Interface	0.2.3	⊕	↻	⊗
☐	crayon	Colored Terminal Output	1.3.4	⊕	↻	⊗
☐	crosstalk	Inter-Widget Interactivity for HTML Widgets	1.1.0.1	⊕	↻	⊗
☐	curl	A Modern and Flexible Web Client for R	4.3	⊕	↻	⊗
☐	cyclocomp	Cyclomatic Complexity of R Code	1.1.0	⊕	↻	⊗
☐	data.table	Extension of 'data.frame'	1.13.2	⊕	↻	⊗



**A better
Shiny app**



Improved project structure

```
1 .
2 +-- config.yml
3 +-- data
4 |   \-- electricity_production.csv
5 +-- dev
6 |   \-- scr_create_energy_chart.R
7 +-- global.R
8 +-- R
9 |   +-- fct_get_usa_energy_production.R
10 |   +-- mod_energy_plot_server.R
11 |   \-- mod_energy_plot_ui.R
12 +-- README.md
13 +-- renv
14 |   +-- activate.R
15 |   +-- library
16 |   |   \-- R-4.0
17 |   |   \-- x86_64-w64-mingw32
18 |   +-- settings.dcf
19 |   \-- staging
20 +-- renv.lock
21 +-- server.R
22 +-- shiny-good-app.Rproj
23 +-- ui.R
24 \-- www
```



Improved project structure

- `config.yml` -> a configuration file
- `data/` -> data for the project
- `dev/` -> scripts and utility tools
- `R/` -> functions and modules
- `renv/` -> local library
- `renv.lock` -> versions of the adopted packages
- `global.r`, `server.r`, `ui.r` -> files of the Shiny app
- `www` -> javascript and css for the Shiny app



Improved code design

```
1 server <- function(input, output, session) {  
2  
3   data_set <- get_usa_energy_production(CONFIG$quandl_key)  
4  
5   data_set_filtered <- reactive({  
6     data_set %>% filter(date > lubridate::ymd(paste0(input$start_year, '-01-01')))  
7   })  
8  
9   long_data_set <- reactive(  
10    pivot_longer(data_set_filtered(), !date, names_to = 'source', values_to = 'production')  
11  )  
12  
13  output$energy_summary <- renderPrint(  
14    summary(data_set_filtered())  
15  )  
16  
17  energy_plot_server('energy_plot', long_data_set)  
18 }
```



Modules

```
1 energy_plot_server <- function(id, data_set){
2   moduleServer(
3     id,
4     function(input, output, session){
5       output$energy_plot <- renderPlotly({
6         plot_ly(data_set(), x=~date, y=~production, color=~source) %>%
7         add_lines() %>%
8         layout(xaxis = list(title='Date'), yaxis = list(title='Production [GWh]'))
9       })
10    }
11  )
12 }
13
```



Functions

```
1 #' Get USA energy production
2 #'
3 #' Retrieve data about the power production from different sources from Quandl APIs
4 #'
5 #' @param quandl_api_key a string with the Quandl API key
6 #'
7 #' @return a data.frame with the yearly time series of USA energy production
8 #' @export
9 #'
10 get_usa_energy_production <- function(quandl_api_key) {
11   Quandl.api_key(api_key = quandl_api_key)
12
13   Quandl::Quandl(c('UENG/EN_USA', 'UENG/ES_USA', 'UENG/ET_PROD_USA', 'UENG/EW_PROD_USA')) %>%
14     rename(
15       date = Date,
16       nuclear = `UENG.EN_USA - Electricity - total nuclear production (Kilowatt-hours, million)`,
17       solar = `UENG.ES_USA - Electricity - total solar production (Kilowatt-hours, million)`,
18       thermal = `UENG.ET_PROD_USA - Total thermal production (kilowatt-hours; million)`,
19       wind = `UENG.EW_PROD_USA - Total wind production (kilowatt-hours; million)`
20     )
21 }
```



Configuration

```
1 library(shiny)
2 library(Quandl)
3 library(tidyverse)
4 library(plotly)
5 library(config)
6
7 CONFIG <- config::get()
```



Improved code design

- Package config to manage configuration
- Roxygen documentation on functions
- Application logic in `global`, `server`, `ui` and in modules
- Business logic in functions
- Prevent useless reactivity on data sourcing
- Tidyverse-like styling and formatting with `lintr`
- Functions and modules to segregate responsibilities



Dependency management

- The renv package creates a local library for each project, in order to let different projects depend on different versions of the same package
- The file `renv.lock` contains the versions – and must be put under version control
- The folder `renv/` contains the packages themselves
- A system-level cache is used not to download the same package twice



renv at a glance

- `renv::init()` -> set up the required files and folders
- `renv::snapshot()` -> freeze into `renv.lock` the installed packages
- `renv::restore()` -> install all packages listed in `renv.lock`

A large, dark, irregular ink blot with splatters on a white background. The blot is composed of several overlapping dark shapes, with a gradient from black to dark blue. It has a rough, textured appearance with many small droplets and splatters radiating outwards, particularly towards the top and right edges.

The next level







Resources

Great resources

Some articles...

<https://shiny.rstudio.com/articles/>

... and a great book to start:

[H. Wickham, *Mastering Shiny*](#)

A great book for advanced topics:

[C. Fay et al, *Engineering Production-Grade Shiny Apps*](#)

The documentation for renv is nice:

<https://rstudio.github.io/renv/index.html>





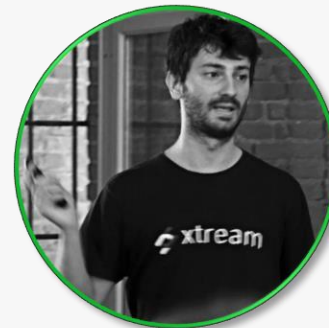
xtream

Web: xtreamers.io

Instagram: [@xtream_srl](https://www.instagram.com/xtream_srl)

LinkedIn: [@xtream-srl](https://www.linkedin.com/company/xtream-srl)

GitHub: [@xtreamsrl](https://github.com/xtreamsrl)



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